9. Air-Cleaning Devices

In the previous chapter, the general methods available for use in cleaning air were discussed. These methods have been applied by engineers to design specific air-cleaning equipment. Generally, a particular air-cleaning device will use more than one of the general methods for cleaning air. In this chapter, the various types of air-cleaning devices that are available will be discussed in some detail.

Air Cleaners for Particulate Contaminants--Mechanical Separators

Mechanical methods for separating particulate matter from the air primarily involve the use of the force of gravity, centrifugal force, and/or impaction. There are two factors that have a primary effect on the collection of particulate matter by mechanical means.

1. The force of gravity or particle inertia increases with the square of the particle diameter.
2. Since particles must travel across the air stream to be collected, the smaller the distance of travel across the air stream, the better the removal of the particles.

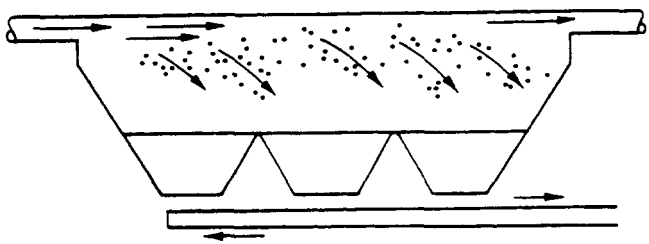
Thus, it is evident from the above that larger particles will be collected more easily by mechanical means. In addition, if the air stream can be made narrower, then the distance of travel for a particle will be less, and particles will be collected more easily.

Gravity Settling Chamber. The simplest type of air cleaner for particulates is the gravity settling chamber. The gravity settling chamber is simply a large settling container attached to the ventilation duct system. As the air enters the large chamber, the velocity slows to below the transport velocity for the particulate matter being carried in the air. Particles settle out of the air into the hopper as a result of the gravitational force acting on the particles. Theoretically, all particles except those affected by Brownian motion (approximately 0.5 micrometers) can be removed by gravity if the settling area can be made large enough. Practically, the settling chamber size must be limited; and, therefore, gravitational settling is effective only for relatively large particles.

Gravity settling chambers may include baffles upon which particulate matter with horizontal inertia will impact. The particles are then attracted by gravity downward into the collection hopper. After the particles are collected, removal may be accomplished by either intermittent cleaning of the hopper or a continuous cleanout using a conveyor system.

Figure 2.9.1

Settling chamber.



Certain hoods and ventilation duct systems have plenums that not only slow the air but also act as settling chambers. For this reason, it is necessary to provide a cleanout in any plenums to remove any settled particulates. In addition, long runs of ducts operating at less than the transport velocity of the particles also act as settling chambers and require the addition of cleanouts.

The efficiency of gravitational settling is difficult to determine because of the turbulence within the chamber. A theoretical formula has been developed to determine a theoretical efficiency for the chamber (Stokes Law). This formula can be useful in estimating the size chamber to be used for a particular minimum particle size. The formula is:

$$D_p = \frac{18\mu Hv}{gL(P_p - \rho)}$$

where

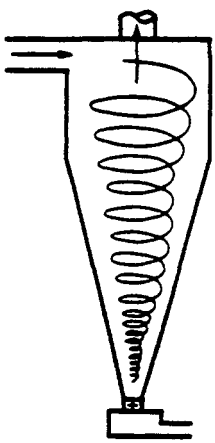
D_p = minimum particle size collected at 100% efficiency
μ = gas viscosity lb/ft-sec
H = chamber height in ft
v = gas velocity in ft/sec
g = 32.2 ft^2/sec
L = chamber length in ft
P_p = particle density in lbs/ft^3
ρ = gas density in lb/ft^3

In general, gravity settling chambers are used to collect large particles, normally in excess of 100 micrometers. They are often used as chip traps in conjunction with woodworking equipment; and in some cases, gravity settling chambers are attached to the system to act as a precleaner to remove large particles before further cleaning of the small particles by another method. Because of the required size and the lack of effectiveness of this method for removal of small particles, the gravity settling chamber is limited in its application as a primary cleaner.

Cyclone Cleaners. The cyclone cleaner represents a second type of mechanical separator. In the cyclone cleaner, the air enters a circular or conical chamber where it swirls around the outside perimeter of the chamber. Particles in the air are acted upon by centrifugal force and forced to the outside walls of the chamber. The particles follow the walls downward to the point where the air reverses flow 180 degrees. At this point, the particles fall into a hopper, and the clean air moves in a vortex upward inside the outer vortex, exiting at the top of the chamber.

Figure 2.9.2

Cyclone.



Cyclones have no moving parts. Special inlets are designed to eliminate re-entrainment of dust from the entry air into that air which is exiting the cyclone. Special materials can be used in a cyclone if the air contains corrosive substances. Water may be sprayed along the wall of the cyclone to increase the capture of particles. The design of a cyclone may or may not be conical. In addition, the cyclone may be horizontal rather than vertical. Also, varying inlet and outlet designs have been used by various cyclone manufacturers.

Efficiency curves are developed by the manufacturers for a particular cyclone. This efficiency is determined from performance tests on the cyclone. There are a number of methods to increase the efficiency of a given cyclone. A higher inlet velocity will increase efficiency since the particles are subjected to a greater centrifugal force. The diameter of the cyclone affects efficiency. The smaller the diameter, the higher the efficiency. High-efficiency cyclones are designed around this principle and generally have a diameter of less than or equal to 9 inches. High-efficiency cyclones often use multiple tubes with the inlet air being split among the tubes. An increase in the body length of a cyclone also increases the efficiency since

it allows for a longer separation time and lowers the possibility for re-entrainment of dust at the outlet. In addition, the inlet can be designed so that entering air is separated from the clean air being removed (shave off inlet); and directional elbows at entry will direct the contaminated air to the outside perimeter of the cyclone, thus separating the inlet and outlet streams.

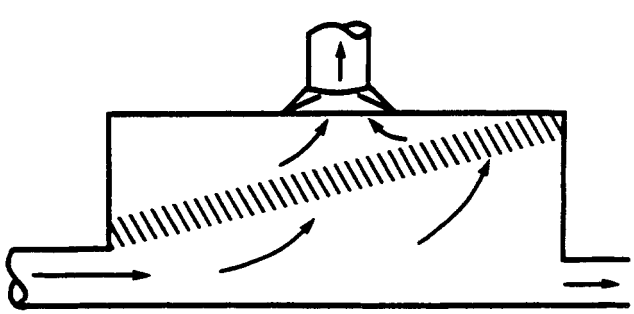
Conventional cyclones are widely used for dust removal in industry. These cyclones have low pressure drops, in the neighborhood of 0.5 to 1.5 inches wg. Cyclones are inexpensive to install and operate; they are efficient for particles in the range of 20 to 40 micrometers; they can be used in high-temperature air streams. And cyclones are basically trouble free, with problems being encountered only where particularly corrosive materials are used or where plugging of the outlet or caking material on the walls result from handling air with a high moisture content.

Small high-efficiency cyclones exhibit pressure drops up to 6 inches of water. These cyclones are used for separation of particles with a diameter of 10 micrometers or more.

Impingement or Impaction Devices. In impingement or impaction devices, air is made to pass around or through specially shaped obstacles or openings. The openings or baffles are widely spaced (generally one inch or more). As the air passes around the obstacle, particle inertia causes the dust to travel in a straight line and hit the obstacle. The dust particles are deflected from the air path and into a dust collection hopper. The particles are then removed either intermittently or continuously from the hopper. The clean air flows out of the collector. There are a multitude of impaction cleaner designs. Some are simple zigzag baffle types, in which the air hits the zigzag baffles and passes upward vertically while the particles are directed downward by the force of gravity and their directional inertia. Another type is circular, with louvers to allow the air to pass through while the particles impact on the louvers and are directed away from the air flow. A third type uses a high-velocity jet caused by passing the air through perforations or nozzles. The air impinges on a plate which is placed in the path of the jet.

Figure 2.9.3

Impaction separator.



The high jet velocities increase the efficiency of such cleaners. In some cases, impaction separators use water as the impingement medium.

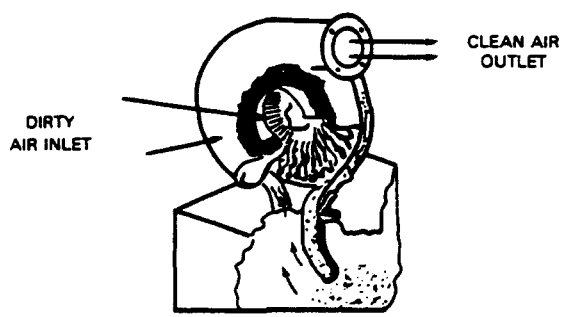
Impaction separators generally operate at a low efficiency (50 to 80 percent). The efficiency of an impaction separator can be increased by lengthening the baffles, increasing the airflow rate, or spacing the baffles closer together. A limiting factor in increasing the efficiency is the potential plugging of the openings and the high pressure loss that may be encountered if small openings or nozzles are used.

Impactors generally have a pressure loss of from 0.5 inches wg to 3 inches wg. They are effective in collecting particles of a diameter greater than 20 micrometers. Other than the problem of plugging, impactors are generally trouble free and can be used in situations where cyclones are applicable.

Dynamic Collectors. The dynamic collector utilizes a fan or impeller that operates in the air stream. The blades gather the dust through centrifugal force, and the blade shape separates the particles from the air.

Figure 2.9.4

Dynamic precipitator.



The efficiency of a dynamic collector is determined by performance tests. Generally, the efficiency falls between that of the conventional cyclone and the high-efficiency cyclone. The efficiency of a dynamic collector remains essentially constant for a given collector over the entire range of operating velocities.

There is no pressure loss present when using a dynamic collector, since the collector acts in the same way as a fan. The dynamic collector is effective for collecting particles of a diameter greater than 10 micrometers. In a corrosive or abrasive atmosphere, wear on the fan or impeller becomes a problem. If the temperature of the carrier gas is in excess of 700°F, specially designed collectors must be used.

Air Cleaners for Particulate Contaminants--Filters

A filter is a porous material composed of granular or fibrous materials that removes particles from a fluid or gas which passes through it. As the fluid flows through the filter medium, particles impact or are intercepted by the filter fibers. Particles build up on the filter and fill void spaces. As the buildup occurs, resistance to the flow increases; and the filter can trap smaller particles. As the filter loads, the resistance increases, and it becomes necessary to replace or clean the filter to maintain the desired air flow. The best filter is one that can efficiently capture and hold particulates with a minimum resistance to flow.

Most filters utilize interception and impacting mechanisms in the filtering process. High-efficiency filters also utilize diffusion to remove particles. Depending upon the filter medium and the particulates involved, electrostatic attraction may also come into play. As the particles build up in the void spaces of the filter, a sieving action becomes important.

Deep-Bed or Mat Filters. Deep-bed or mat filters are made up of beds of granular or fibrous materials arranged to provide a porous cross section (90 to 99 percent void space). As a result, these filters have a low resistance and high dust load storage capacity. Generally such filters are cleaned by replacing.

These filters are inexpensive to buy and can be designed for high temperatures. In addition, such filters can be used where corrosive contaminants are present. However, this type filter is not widely used in air pollution control since most filters of this type cannot be cleaned. In some cases, mat filters have been used for filtration of sulfuric or phosphoric acid (coke-box type filters) and to collect dust and fumes from open-hearth furnaces. The deep-bed or mat filter can be graduated in order to obtain higher efficiencies with an accompanying increase in resistance to flow. Some special-design filters of this type have been used as moving filter beds or self-cleaning mat filters.

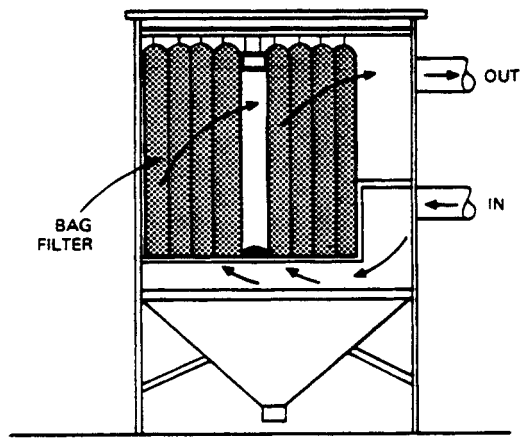
Fabric Filters--Bag Houses. In a bag house, the air flows into the bottom of a fabric tube; and, as it passes through the fabric, the particulate matter is removed and builds up on the fabric. Efficiency of such filters is high (up to 99 percent) for small particles of 0.25 micrometers or more. The particulate buildup (cake) provides a sieving action for particulate matter in subsequently cleaned air. As the cake forms, the resistance to flow increases, ultimately resulting in the need to clean the filter.

Typically, the resistance to flow encountered with the fabric type filter varies between 3 to 8 inches wg. Resistance to flow in excess of the upper limit for a particular filter indicates the need for cleaning.

There are three basic categories of bag houses being used in industry. These categories are based upon the method that is employed for cleaning the filters. The intermittently cleaned bag house is cleaned by shutting down the filtering operation and removing the cake buildup from the filters, generally

Figure 2.9.5

Bag house.



by shaking or rapping the filters. After the filters have been cleaned, the bag house can be put back into operation. Such a system requires either that the process be shut down while the cleaning takes place or that redundant cleaners be installed. Periodically cleaned equipment is constructed so that separate chambers within the bag house can be shut off from the process and the filters cleaned by rapping or shaking while the other chambers are operating to clean the air. The compartments of a periodically cleaned filter act as essentially separate air cleaners with varying efficiency, flow rate, and resistance. In continuously cleaned filters, the cleaning mechanism involves a continuously moving blow ring which directs a reverse flow of air through the filter to remove the cake. This ring moves up and down the filter tubes while the bag house remains in operation.

The use of a bag house is somewhat limited when high temperatures are encountered. Some fabrics are more heat resistant than others; but in general, the maximum air temperature for which a conventional bag house using cotton filters can be used is approximately 180°F.

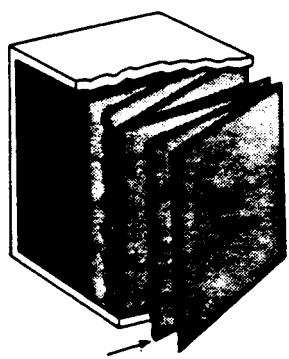
Fabric filters are very widely used for many applications. This type filter is often used to remove dust, fly ash, and such things as electric-furnace fume emissions. However, certain problems can occur with the use of fabric filters. For example, excessive moisture in the air can make the cake difficult to remove. Also, fabric filters are subject to variations in flow between the filter immediately after cleaning and after some period of operation. Thus, a constant level of particulate removal is not likely to be obtained. In addition, fabric filters require large space allocations because of their size.

High-Efficiency Panel Filters. This type filter involves a continuous sheet of filter medium that is accordion folded, with or without separators

between the folds. The high-efficiency panel filter operates at low velocities to capture very small particles in the range of 0.01 micrometers or less by diffusion. In general, the efficiency of capture is in the range of 99.9 percent. The storage capacity of such filters is low because of the high density of the filter medium. Filters of this type are rated by a military standard (Department of Defense Smoke Penetration and Air Resistance Test). The rated resistance for this type of filter is a maximum of 1 inches wg.

Figure 2.9.6

High efficiency filter.



High-efficiency filters are most often used in such cases as clean rooms, food processing plants, and hospitals. In these cases, the dirty filter is disposed of rather than cleaned.

Air Cleaning for Particulate Contaminants--Wet Collectors

Wet collectors, often called scrubbers, involve the introduction of a liquid to the air stream to remove particulate matter. Wet collectors employ impaction as their primary method of air cleaning. The particles and the water droplets collide, and the liquid condenses on the particle, increasing its size and density. Small droplets provide a maximum area for impingement. However, the impingement efficiency will fall off if the droplets become too small, thus allowing the particle to pass between them. Small particles also diffuse to the droplets within a wet scrubber. Humidification of the air may cause some of the small particles to flocculate, thus increasing the efficiency of a wet scrubber.

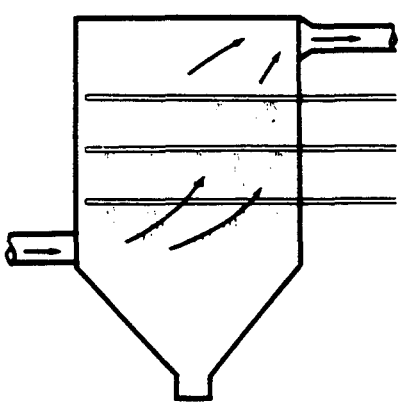
Wet scrubbers normally result in high-efficiency air cleaning with low initial cost. In general, however, the efficiency obtained is directly related to the resistance (pressure drop) encountered. Wet scrubbers are effective in high temperatures or corrosive atmospheres. Removal of the waste sludge from a wet scrubber can present a problem and must be considered in their installation. The maintenance problems involved with wet scrubbing vary, depending upon the type of scrubber and the design.

Chamber Scrubbers--Spray Towers. In a spray tower, water is sprayed into a chamber through which air is being passed. Baffles may be present to direct the air in order to increase the time within the chamber and, thus, the potential for contact between the particles and the water droplets. The spray tower is generally effective for particles of a diameter of 10 micrometers or more. High-pressure sprays can be used to capture particles in the one micrometer range.

The simplest spray tower design involves a gravity-chamber spray with the air being introduced at the bottom of the chamber. This design results in a very simple, trouble-free operation with water being introduced at line-pressure rates. In general, the spray tower operates with a low pressure drop of from 1 to 1.5 inches wg.

Figure 2.9.7

Spray tower.



The conventional spray tower operates at a relatively low efficiency of collection. The high-efficiency type (fog tower) operates at a significantly higher efficiency and is, in fact, one of the most efficient of the wet collectors. However, the fog tower does require significant quantities of water and has an increased resistance to air flow of from 2 to 4 inches wg.

Cyclonic Scrubbers (Wet Cyclones). The cyclonic scrubber is similar to the normal dry cyclone with a water spray introduced tangential to the air entry. The cyclone action removes the particles that are wetted by the spray. This type cleaner is effective for particles of 5 micrometers with an efficiency of greater than 90 percent. The pressure drop is from 2 to 6 inches wg. Special designs have been used employing vanes and various nozzle arrangements to improve the efficiency of such scrubbers.

Self-Induced Spray Scrubbers. In this type of air cleaner, the air stream passes through constrictions or nozzles to increase its velocity. The

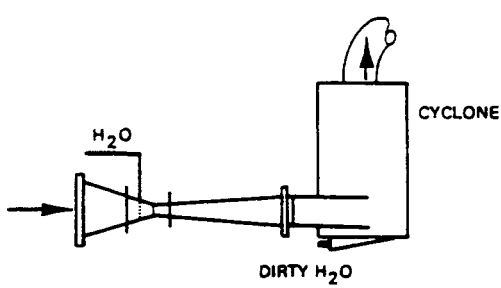
high-velocity air stream then impinges on a liquid. The liquid is fragmented by the force of the air, and a spray is created. The droplets and particles impact in the turbulent spray. The droplets and particles then impact on baffles for removal. The efficiency of such a system is near 90 percent for particles of 2 micrometers in diameter. The pressure drop of such a scrubber is in the range of 3 to 6 inches wg.

Wet Impingement Scrubbers. In a wet impingement scrubber, the air enters a spray area. The droplets in the spray capture the particles, which then impact on the baffle plate. Small holes in the baffle plate allow the air to pass. The efficiency of such a scrubber is approximately 90 percent for particles with a diameter of 2 micrometers. The pressure drop in such systems is from 1.5 to 8 inches of water, depending upon design.

Venturi Scrubber. In the venturi scrubber, air passes through a venturi (constriction). Liquid is introduced perpendicular to the air flow at or before the venturi. The liquid is atomized by the force of the air, and turbulence causes a mixing of the particles and liquid. The particles impact upon the spray drops and are separated from the air stream. The efficiency of such scrubbers can be high (99 percent) for submicrometer particles. However, for such units the pressure drop is also very high, with some scrubbers operating at 60 to 100 inches wg.

Figure 2.9.8

Venturi scrubber.



A venturi scrubber is often used in conjunction with a cyclone separator which collects the droplets of water holding the particles. Small particles passing through the venturi impact, impinge, or diffuse on the water droplets, thus increasing the size of the particle (droplet and particle) to be removed. The cyclone can then be effective in removing these larger particles at relatively high efficiencies.

The effectiveness of the venturi for removing small particles at high efficiency has led to its use in difficult air-cleaning jobs. However, as with other cleaners, this increase in cleaning efficiency is not without cost. The high pressure drop that occurs using a venturi results in the need for large and costly air moving equipment.

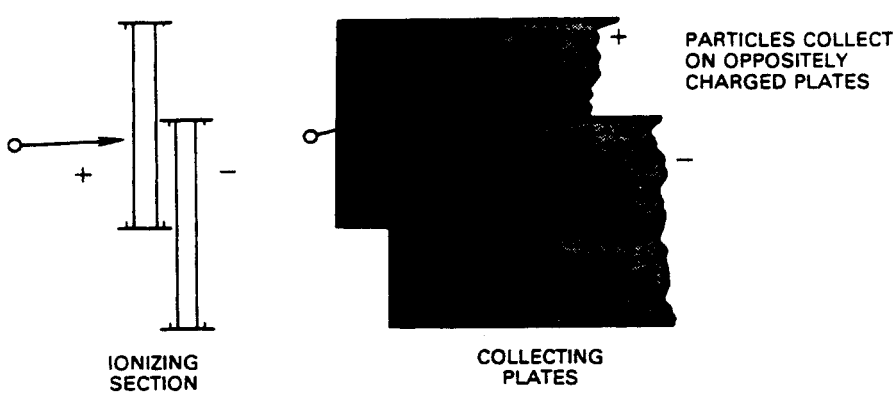
Mechanical Scrubbers. The mechanical scrubber involves the use of a rotor propeller or fan to break up a liquid. The turbulence caused by the liquid breakup and the movement of the air enhance the potential contact of the particles and the droplets. The droplets and particles then fall into the liquid to be removed. This type of scrubber has an efficiency of approximately 90 percent for submicrometer particles. No resistance is present for this type of cleaner since the scrubber acts as a fan. However, the cost of operation can be high, because of the power requirements of the scrubber itself.

Air Cleaning for Particulates--Electrostatic Precipitators

Electrostatic precipitation involves charging the particles in the air with an electrical charge that is opposite to the charge applied to the plates where the particles are to be gathered. A high-voltage discharge electrode is placed opposite a grounded electrode. The gas passes through the electrodes and becomes ionized. (This is called the "corona.") The ions attach to the particles and are attracted to the oppositely charged plates, where they gather. The dust layer must not build up on the plates, or the efficiency of the precipitator will be affected. In some designs, the collector plates are shielded to prevent re-entrainment of the particles as air blows by the plates. The dust is cleaned from the plates by mechanical rapping or vibrating or by a liquid film on the plates themselves. Cleaning without re-entrainment of dust particles presents a difficult design problem. In some cases, shielded collector plates are used to lower the possibility of re-entrainment. Precipitators can also be sectionalized to facilitate cleaning without re-entry of dust. The liquid film type is particularly effective, as the dust is continuously washed from the plates with virtually no re-entrainment occurring.

Figure 2.9.9

Electrostatic precipitator.



Once the plates have been cleaned, it is necessary to gather the removed dust and dispose of it. Baffled hoppers are used to collect the dust. The baffles act to prevent dirty air from bypassing the collection plates. The dust collected in the hopper can be removed intermittently by a conveying system.

There are basically two types of electrostatic precipitators. The first of these is a low-voltage precipitator that operates at approximately 12,000 volts. In low-voltage precipitators, the ionizer and collecting sections are separated. This type of precipitator is approximately 95 to 99 percent efficient in the removal of oil fumes. Except for this application, low-voltage precipitators are not widely installed in industry.

The second type of precipitator, the high-voltage precipitator, operates at up to 75,000 volts. The collector may be made up of tubes when a wet gas is being processed or plates for the cleaning of dry dust. The high-voltage precipitator is widely used for control of stack effluents from such operations as blast furnaces, power-generating stations, electric furnaces, and cement plants. A high-voltage precipitator is 95 to 99 percent efficient for 0.25 micrometer particles. Electrostatic precipitators exhibit low pressure drops from 0.5 to 1 inches wg. However, the cost of installation of precipitators is high, and space requirements for installation are large.

Air Cleaning for Gases--Adsorption

The previous discussion was limited to cleaning particulate matter from the air. The methods applied are not generally applicable to cleaning gases and vapors from the air. Special methods must be applied for this type of air cleaning.

Adsorption is a physical phenomenon in which the gas or liquid attaches or condenses on the surface of a solid (adsorbent). The forces that act to cause adsorption include van der Waal forces between the molecules and capillary condensation.

The general principles of adsorption involve the passage of air containing a gas or vapor to be removed through a bed of granulated adsorbent. The most common adsorbents are forms of activated carbon. Molecules of the gas attach to the granules of the adsorbent. The process is reversed to remove the adsorbate. Various adsorbents have specific applications for certain gases or vapors. For example, activated carbon has application with hydrocarbon vapors, while silica gel adsorbs water in preference to hydrocarbons.

The time the adsorbate spends in the adsorbent (dwell time) is important. If this time is not sufficient to allow the adsorbing action to take place, the efficiency of the system will suffer. Any velocity through the bed above the critical velocity will result in a decrease in the percentage of capture.

Some general principles apply to the process of adsorption. These principles are:

1. Generally, adsorption varies directly with the vapor pressure of the adsorbate.

2. Adsorption varies inversely with the temperature and directly with the pressure of the system.
3. The more dense the adsorbate, the greater the quantity adsorbed.
4. When using activated carbon as an adsorbent, the material with the greatest surface area and the lowest bulk density for a given size (mesh) is the most efficient and holds the most adsorbate.
5. The ease of adsorption increases as the boiling point of the gas or vapor increases.
6. For organic vapors, a larger number of carbon atoms per molecule increases the ease of adsorption.
7. C_1 (carbon) and C_2 compounds (with one or two carbon atoms per molecule) do not adsorb.
8. Unsaturated compounds of carbon adsorb better than saturated compounds of carbon.
9. Synthetic hydrosilicates are more efficient in capture than activated carbon, but do not have the holding capacity.
(Adapted from Air Pollution Manual, Part II, AIHA, 1968.)

Recirculating Adsorbing Systems. Recirculating adsorbing systems consist of a thin, loosely compacted bed approximately one-half inch in depth. The dwell time in such a bed is generally 0.05 seconds or less. The resistance to flow is less than 0.25 inches wg. The retention of such beds is between 5 and 50 percent. Generally such systems operate as one-pass systems and are used most often in room air conditioning. These systems are not suitable for general air pollution work.

One-Pass Nonregenerative Systems. These systems are made up of beds of three-quarters to several inches thick. Dwell time varies from 0.075 to 0.5 seconds. Such systems operate effectively on concentrations of a few parts per million. The pressure drop through such a bed is generally 0.5 inches wg. These systems are used where a small amount of contaminant is present.

One-Pass Regenerative Systems. The one-pass regenerative system is used when the contaminant exceeds a few parts per million. Two adsorber beds are used; one operates while the other is being regenerated. The beds in such systems are generally one to three feet in depth with a dwell time of 0.6 to 6.0 seconds. The pressure drop experienced in such a system is from 3 to 15 inches wg. Regeneration is accomplished by applying low pressure steam to raise the bed temperature and drive out the adsorbent. This type of system is widely used in industry.

Air Cleaning for Gases--Absorption

In the process of absorption, gas is dissolved in a liquid that is mixed with the gas. This process is commonly called "gas scrubbing" in the air pollution control industry.

In absorption, the air containing the contaminant gas or vapor is introduced to a liquid. The liquid dissolves the gas or vapor. The combined liquid-contaminant waste is then removed. Turbulent mixing of the liquid and gas increases the possibility of dissolving the gas in the liquid.

Among the types of absorption cleaners currently used in industry are packed towers, plate towers, and spray towers. A packed tower is filled with various shaped metals or crushed rock. Air enters the tower at the bottom, and liquid enters at the top. The liquid normally used is water, though other special liquids may be used for removal of a particular gas or vapor. The gas to be removed is dissolved in the liquid and flows out at the bottom of the tower. Air velocities are normally in the range of one foot per second for a packed tower. Packed towers can present problems with flooding when the air velocity is significantly greater and causes the liquid to bubble up. Particles can clog the packing medium. Generally, packed towers are used in small installations where corrosive materials exist.

Plate towers have plates with holes or slots in them. Air bubbles up in the down-flowing liquid, and the contaminating gas is absorbed. Air velocities in plate towers generally are in the area of one to two feet per second. These towers are applicable where higher velocities of air are required and are less susceptible to flooding and plugging. Cooling coils can be installed to cool the liquid and thus cool the incoming air.

The spray tower is similar to the spray tower used in particulate removal. A downward spray of liquid interacts with the air moving upward through the tower. Often the air is moving in a cyclonic motion. Spray towers exhibit relatively low pressure drop and are best used with easily absorbed gases. Sometimes a venturi is included in such a tower to maximize the mixing. These towers are particularly useful where dust particles are to be removed in addition to a gas or vapor.

There are other special types of absorbing equipment available. These include jet scrubbers, wet cell washers, and wet scrubbers that are normally designed for particulates, but which can be used to remove gases or vapors.

Air Cleaning for Gases--Incinerators

Incineration is rapid high-temperature oxidation of a contaminant gas. Incomplete combustion will normally cause a problem; therefore, the incineration must be controlled and complete.

Combustion requires the following three "t's":

A. Time (reaction time)
B. Temperature (increases reaction rate)
C. Turbulence (to mix oxygen, fuel, and contaminant).

Fuel and air are mixed and passed to a combustion chamber. The flame in the combustion chamber burns the contaminant. A high temperature is present as a result of the combustion (1500°F will destroy any odorous organic material or odorous aerosol). The range of concentration must be maintained outside the range of the LEL and UEL to avoid spontaneous explosions. In some cases, a waste liquid can be burned if it has been atomized by spraying prior to combustion.

Incineration can cause a number of problems and requires safety precautions. Excessive temperature of the air after burning can be controlled by adding dilution air downstream from the combustion. Flashback can be controlled by diluting the incoming gases below the LEL, increasing the velocity of incoming air to a velocity greater than the flame propagation rate, passing the air through plates with openings too small to permit backward flame propagation, or placing a water spray on the incoming side that will stop any combustion. An incineration unit must have an automatic fuel shutoff to prevent problems occurring because of flame failure.

Air Cleaning for Gases--Catalytic Combustion (Oxidation)

Often it is necessary to provide a high temperature for complete combustion of the contaminant. In such cases, excessive cost is experienced both in terms of fuel use and the precautions necessary to maintain the incinerator itself. This situation can be overcome by the use of catalytic combustion.

In catalytic combustion, the contaminated air is passed through a bed containing a catalyst. A commonly used catalyst is an alloy of platinum that is placed on wires to form a mesh, or rods, beads, pellets, etc. The catalyst bed must expose the maximum surface of the catalyst to the gas stream. Sufficient area must be present to complete the oxidation. Gases may be heated prior to the catalytic oxidation. Catalytic combustion results in a low pressure drop, usually 1/4 to 1/2 inches wg. Catalytic combustion is not suitable for air containing fly ash, inorganic solids, vaporized metals, and some halogenated hydrocarbons.

Summary

Cleaning contaminants from exhaust air involves the removal of particulate matter, gases, and vapors. General methods for cleaning the air of these contaminants differ from each contaminant type. Various types of air cleaners have been designed to accomplish the desired cleaning. The cleaners operate at different efficiencies in removing particular contaminants. In general, as the efficiency of the cleaner increases, the cost of installation and operation also increases. For particulate cleaners, the resistance to flow increases in direct relationship to an increased efficiency. This in turn necessitates the use of more powerful air movers, thus increasing the cost of operation. Before selecting the appropriate air cleaner, the industrial hygiene engineer must be familiar with the process requirements and the desired quality of the exit air.

Table 2.9.1 summarizes the most common types of air cleaners discussed in this chapter.

Table 2.9.1

Common types of air cleaners.

Type	Effective Range (micrometers)	Pressure Loss (in. wg.)	Space	Problems	Common Use Areas
Particulate Cleaners					
Dry Type					
Settling Chamber	>100	Varies with size and design	Large	Only useful as pre-cleaner to remove large particles	Woodworking chip traps
Dry Cyclone	20-40	1 - 1.5	Large	Corrosive or moisture-laden contaminants	Crushing and grinding, metal working, woodworking
High Efficiency Cyclone	≥10	3 - 6	Moderate	Corrosive or moisture-laden contaminants	Crushing and grinding, fly ash, drying, metal working
Impingement	>20	0.5 - 3	Small	Low efficiency; corrosive or moisture-laden air	Fly ash, crushing and grinding, cement clinker coolers
Dynamic Collectors	>10	None*	Small	Corrosive or moisture-laden air	Metal working, woodworking
Mat Filters	4-20	2 - 8	Small	Noncleanable	Coke box, open hearth furnaces
Bag Houses	≥0.25	3 - 6	Large	High temperature, moisture, cleaning	Crushing and grinding, material handling, foundries, wood-working, fly ash
High Efficiency Filters	≥0.01	1	Small	Low velocity of air flow, low capacity	Clean rooms, food processing, hospitals
Electrostatic Precipitators	≥0.25	0.5 - 1	Large	Installation cost high	Fly ash, blast and open hearth furnaces
Wet Type					
Spray Towers	≥10	1 - 1.5	Large	Clogging, low efficiency	Crushing and grinding, material handling, foundries, drying, metal working

Table 2.9.1 (continued)

Type	Effective Range (micrometers)	Pressure Loss (in. wg.)	Space	Problems	Common Use Areas
Wet Type-Continued					
Wet Cyclone	≥5	2 - 6	Moderate	Water usage	Same as above
Self-Induced Spray	≥2	3 - 6	Small	Water usage	Same as above
Wet Impingement	≥2	1.5 - 8	Small	Water usage	Same as above
Venturi	≥0.5	60 - 100	Moderate	Water usage, high power requirements	Where high efficiency cleaning for small particles is required
Mechanical	≥0.5	None	Small	High power requirements	Same as venturi
Gases and Vapors					
Adsorption--One Pass Regenerative	--	3 - 15	Large	Contamination of adsorbent	Solvent recovery, odor control, chemical manufacturing
Absorption--(Spray and Packed Tower)	--	Varies	Large	Plugging and flooding	Chemical manufacturing
Incineration	--	--	Large	Safety, cost of fuel	Odor control, reducing plume opacity, removal of hydrocarbons
Catalytic Combustion	--	0.5	Moderate	Not applicable where certain types of particulates are present	Organic gases and vapors